Project - Low Level Design on Sports Content generator

Course Name: GenAI

Institution Name: Medicaps University – Datagami Skill Based Course

Team Members:

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Low-Level Design (LLD)
Spors content generator

1. System Overview

A web-based AI platform that:

- Fetches live cricket match data (CricAPI)
- Generates realistic sports articles (Groq LLM)
- Supports user authentication
- Stores article history
- Displays trending live matches (IPL 2026)

2. Architecture (High-Level Flow)

```
User (Browser)
    ↓
Streamlit UI (Frontend + Controller)
    ↓
------------------------------------- |
Auth Layer (Login/Register)    |
|  Article Service (Groq API)    |
|  Match Service (CricAPI)       |
|  History Service (SQLite DB) |
-------------------------------------
    ↓
External APIs + Database
```

3. Modules & Responsibilities

3.1 UI Layer (Streamlit) Components:

- Sidebar Navigation
- Generate Article Page
- History Page
- Trending Matches Column (Right side)

Responsibilities:

- Handle user input
- Render UI
- Manage session state
- Trigger backend functions

3.2 Authentication Module Functions:

register_user(username, password)

login_user(username, password) Logic:

- Password hashing using bcrypt
- Store credentials in SQLite
- Validate login credentials

3.3Match Data Service (CricAPI) Function:

get_match_data() Flow:

Call CricAPI → Parse JSON → Extract top 3 matches → Format string Output:

Match: India vs Australia
Status: Live Venue:
Mumbai

3.4 AI Article Generator (Groq LLM) Function:

generate_article(query, match_data, temperature)
Flow:

User Query + Match Data
↓
Prompt Engineering
↓
Groq API (Llama-3.3-70B)
↓
Generated Article

Prompt Strategy:

- Role: Sports Journalist
- Context: Live match data
- Output: ESPN/Cricbuzz style article

3.5 History Service (SQLite) Tables:

Users Table

username TEXT PRIMARY KEY

password TEXT History Table

id INTEGER PRIMARY KEY
username TEXT query TEXT
article TEXT timestamp
TEXT Functions:

save_history(user, query, article)
load_history(user)

3.6 Session Management Using

Streamlit: st.session_state.user

st.session_state.page States:

- user = None → Not logged in
- page = generate | history

4. UI Design (Detailed)

4.1 Sidebar

Sports AI Journalism

Navigation
[History]
[Home]

Logged in as: user
[Logout]

4.2 Main Layout

Using:

col1, col2 = st.columns([3,1]) Left

Column:

- Input box
- Generate button
- Article output Right Column:

Trending
Live IPL Match 2026
Match details (from API)

5. Data Flow

Article Generation Flow

User enters topic
↓
Click Generate
↓
Fetch match data (CricAPI)
↓
Send prompt to Groq
↓
Receive article
↓
Display + Save in DB

History Flow

User clicks History
↓

Fetch records from SQLite
↓
Display using expanders

6. Error Handling

Component	Handling
CricAPI	Check status code
Groq API	Retry / limit
DB	Try-catch on insert
Input	Empty query warning

7. Key Design Decisions

1. Streamlit as Full Stack

- UI + Controller in one file
- Faster development

2. SQLite

- Lightweight
- No external DB setup

3. Groq API

- Fast inference (Llama-3)
- Real-time generation

4. No Dataset

- Uses live API instead
- More realistic outputs

8. Scalability Considerations

Future improvements:

- Replace SQLite → PostgreSQL
- Add Redis caching for match data
- Use FastAPI backend
- Deploy on AWS/GCP

9. Folder Structure

```
sports-ai/
│
├── app.py          # Main app (UI + logic)
├── .env            # API keys
├── sports_ai.db    # SQLite DB
├── requirements.txt
```

10. Tech Stack

Layer	Technology
Frontend	Streamlit
Backend	Python
LLM	Groq (Llama-3.3)
Sports API	CricAPI
Layer	Technology
Database	SQLite

11. Final Summary

The Sports Content Generator is a scalable and modular GenAI system that effectively combines:

- Real-time sports data
- Context-aware retrieval (RAG)
- Advanced language models

This design ensures high-quality, dynamic, and user-relevant sports content generation.